Awab Qureshi

awabq.com | linkedin.com/in/awabq | github.com/aaqsr | awab.qureshi@uwaterloo.ca

Skills

Languages C, C++, Rust, Python, System Verilog, JavaScript/TypeScript, SQL, x86_64 Assembly, Bash, LISP
Technologies Git, Linux, Kafka, Redis, OpenGL, AWS, Numpy/Pandas, PostgreSQL, Fusion 360 CAD, AMD Vivado, Docker

Experience

GTS (Global Trading Systems) | *C++ Developer Co-op* — *May 2026 – Sep 2026*

- Rebuilt a legacy, never-deployed data reconciliation engine into a **production-ready service**, able to reconcile the firm's financial order-log data from two **Kafka** sources in **real-time** at **~100K** messages/sec.
- Wrote unit and integration tests, **extending coverage** to the engine's core matching, recovery, and output paths for the first time, **growing the test suite 3.8×** and surfacing **critical data races and use-after-frees** that were then fixed.
- Profiled the service with **Perfetto**, then rewrote its contended hot path with Asio **coroutines**, cutting it **from 5.76s to 95ms**.
- Replaced rigid config with a grammar over **arbitrary FlatBuffer schemas**, letting ops add new recon jobs **without writing code**.
- Revived an unowned C# operations **dashboard** used firm-wide daily with a data-diff view, using custom **agentic LLM** tooling.

Huawei | *Software Engineering Co-op* — *Sep 2025 – Jan 2026*

- Researched **compiler optimisations**, achieving a **12.6% reduction** in binary size and a **3.7% improvement** in execution latency.
- Implemented a **kernel-level** regex engine integrated into a **Linux Security Module** to enforce fine-grained file access control.
- Designed and built a **multithreaded, production-ready** user action authorisation service with **high unit test coverage**, utilising **TLS**-secured **Unix Domain Sockets** for **inter-process communication**, and **gRPC** for client communication.

Christie Digital Systems | *Software Engineering Co-op* — *Jan 2025 – May 2025*

- Characterised **computer vision** systems, **automated** manual correction and developed an **optical testing framework** with C++.
- Extended a **critical binary encoder** in C++ and **interfaced directly with projection hardware** to implement multi-colour features.
- Researched **solutions to optical distortions**, utilising Numpy, Pandas and Matplotlib to analyse lab data in **Python**, developing improved **camera calibration** techniques and **subpixel projector alignment**, leading to **80% error reduction** at the optical centre.

University of Waterloo | *Software Engineering Instructional Support Assistant* — *May 2024 – Sep 2024*

- Planned and **taught** weekly tutorials on **modern C++** and **object-oriented design** to over 100 **second-year UW SWE students**.

Expresume | *Full-Stack Web Developer Intern* — *May 2023 – Sep 2023*

- Spearheaded **full-stack design** for the **AI platform**, using **PostgreSQL**, **AWS**, **Tailwind** and **NextJS** to **decrease bundle size by 60%**.

Projects

Fluid and Rigid-Body Simulator [*awabq.com/488*] | *C++, GLSL*

- Developed **real-time, lock-free multithreaded physics engine** in C++ with quaternion-based rotation for **arbitrary convex meshes**.
- Implemented a **water simulation** by discretising shallow water **PDEs** with adaptive time-stepping for numerical stability.
- Coupled fluid engine with the rigid-body engine to enable **fluid-rigid body interaction** enabling realistic buoyancy and splashes.
- Developed custom **OpenGL rendering pipeline** from scratch, with geometry tessellation for **continuous fluid surface rendering**.

Java Compiler [*awabq.com/444*] | *Rust, x86_64 Assembly, Python*

- Built a **Java 1.3 to i386 compiler** in **Rust (~15,000 lines)** with a hand-rolled 7-stage pipeline: lexer, **LALR(1) parser generator** (emitting **4,000+** lines of Rust parsing code), static semantic analyser, type checker, optimiser, and **x86 code generator**.
- Implemented **virtual dispatch**, **inheritance**, **method resolution**, **runtime type checks** and Java's **package import** semantics.
- Built cross-architecture testing infrastructure executing **200+ programs**, rigorously validating against the Java specification.

Activities

Undergrad Mathematics Society of UW | *President* — *Sep 2024 – Jan 2025*

- Acted as the main representative for **over 6,500 math students** and helped strategically **allocate a budget of approx. $100,000**.

Education

University of Waterloo — *Sep 2022 – Apr 2027*

Candidate for *Bachelor of Computer Science, Digital Hardware Specialisation*, GPA: 3.6

Relevant Coursework: Computer Graphics, Compiler Construction, Embedded Software, Digital Hardware Design, Concurrency, Operating Systems, Data Structures & Algorithms, Agile Project Management, Numerical Computing.